Thinking Inside the Box: Efficient Privacy-First Architecture for Distributed Reasoning using Federated Large Language Models

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Abstract

Federated Learning (FL) has emerged as a critical technology for training AI on sensitive data without compromising privacy. However, current architectures face a strict "Privacy-Utility-Efficiency Trilemma," where optimizing for data security often degrades model reasoning capabilities or incurs prohibitive computational costs. This project proposes a novel architecture, "Thinking Inside the Box," which resolves these conventional trade-offs by decoupling **Reasoning** (using Federated Low-Rank Adapters) from **Knowledge** (using local Retrieval Augmented Generation). By simulating a cross-silo network of financial institutions using the IBM Anti-Money Laundering (AML) dataset, this research aims to demonstrate that complex fraud detection can be performed with high utility and efficiency while maintaining absolute data sovereignty.

1 Introduction

1.1 Background and Motivation

Since its inception in 2017, Federated Learning (FL) has transformed how industries leverage sensitive data [1]. By training models locally and sharing only weight updates, FL has enabled breakthrough applications in **Healthcare** (collaborative disease detection across hospitals), **Mobile Computing** (predictive text on smartphones), and **Finance** (inter-bank fraud detection). In the financial sector specifically, FL offers a unique opportunity to combat money laundering by aggregating patterns from multiple banks without violating strict data privacy regulations (e.g., GDPR).

However, deploying Large Language Models (LLMs) in these federated environments introduces a fundamental **Trilemma** [2]:

1. **Privacy:** Keeping data local prevents centralized reasoning.
2. **Utility:** Local models often lack the "global context" required for complex fraud detection.
3. **Efficiency:** Transmitting massive LLM weights overwhelms network bandwidth.

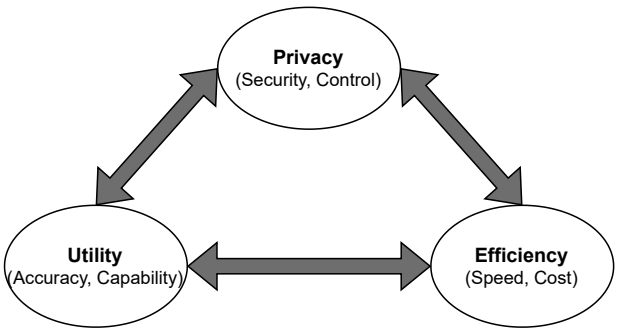

Figure 1: The Federated Learning Trilemma: Traditional architectures force a trade-off. This project aims to **circumvent** this by decoupling reasoning from knowledge.

1.2 Proposed Solution

This project proposes a paradigm shift to **resolve** these conflicting objectives. By decoupling the "Brain" (Reasoning) from the "Memory" (Knowledge), we can train lightweight adapters to perform logic [3] while keeping all sensitive records in a secure, local Retrieval Augmented Generation (RAG) system [4].

1.3 Related Work

The key background sources that enable and justify this novel approach are presented and categorised below.

Table 1: Key References Supporting the Proposed Architecture (Categorized)

Category	Source	Key Concept	Relevance to Project
Theoretical Foundation	Zhou et al. (2023), *DualMask: Federated Optimization of Trilemma [2]*	Formally defines the "Trilemma" in Federated Learning, where privacy, model accuracy, and system efficiency conflict.	Provides the theoretical problem statement that the "Thinking Inside the Box" architecture aims to resolve.
Theoretical Foundation	Allouah et al. (2023), *On the Privacy–Robustness–Utility Trilemma [5]*	Analyzes trade-offs in distributed systems, particularly robustness against adversarial attacks.	Justifies why the architecture (which keeps raw data offline) is resilient to data poisoning and leakage.
Architectural Components	DeepSeek-AI et al. (2025), *DeepSeek-R1: Incentivizing Reasoning Capability [6]*	Introduces "Reasoning Models" that use Chain-of-Thought (CoT) to solve complex logic puzzles rather than just predicting words.	This model class is utilised to enable "local investigation" capabilities that standard LLMs lack.
Architectural Components	Lewis et al. (2020), *Retrieval-Augmented Generation (RAG) [4]*	Establishes the RAG framework for connecting LLMs to external databases.	Basis for the "External Memory" component, allowing secure access to private bank data without training on it.
Efficiency Mechanisms	Hu et al. (2022), *LoRA: Low-Rank Adaptation of LLMs [7]*	Introduces efficient fine-tuning by updating only a small subset of parameters via low-rank decomposition.	Core enabler of the architecture, allowing transmission of tiny "adapters" instead of massive model weights.
Efficiency Mechanisms	Wang et al. (2024), *FLORA: Federated Low-Rank Adaptation [3]*	Extends low-rank adaptation to federated settings, optimizing communication efficiency while maintaining model performance.	Strengthens the deployment strategy by reducing communication overhead and preserving personalisation.

2 Aim and Objectives

2.1 Aim

This project aims to design and evaluate a privacy-preserving architecture integrating **Low-Rank Adaptation (FLoRA)** [3] and **Retrieval Augmented Generation (RAG)** [4]. By decoupling reasoning from knowledge, the system seeks to resolve the "Privacy-Utility-Efficiency Trilemma" in financial forensics, enabling distributed fraud detection on the **IBM AML dataset** [8] without exposing raw transaction data.

2.2 Objectives

To achieve this, the project needs to meet the following objectives:

1. **Develop a Federated Simulation:** Utilise **Flower** [9] to simulate a cross-silo network of banking institutions using the **IBM AML dataset** [8], explicitly modeling realistic heterogeneous (non-IID) data distributions.
2. **Implement Local RAG:** Integrate on-device **Retrieval Augmented Generation** [4] to enable local models to query private transaction logs for reasoning. This ensures enhanced fraud detection capabilities while raw data remains strictly local.
3. **Evaluate via the "Trilemma" Framework:** Benchmark the architecture against baselines across **Utility** (F1-Score [10]), **Efficiency** (Communication Volume, Latency), and **Privacy** (Resilience to data leakage), validating the effective resolution of the trade-off.

3 Planning

3.1 Work Plan

The project follows a linear **technical** execution path, supported by **parallel documentation streams**. This "Simulation-First" strategy ensures that hardware risks are mitigated while writing tasks (Poster and Thesis drafting) occur concurrently with experimental phases. The schedule is aggressively front-loaded to ensure the core technical implementation is completed by March 20th.

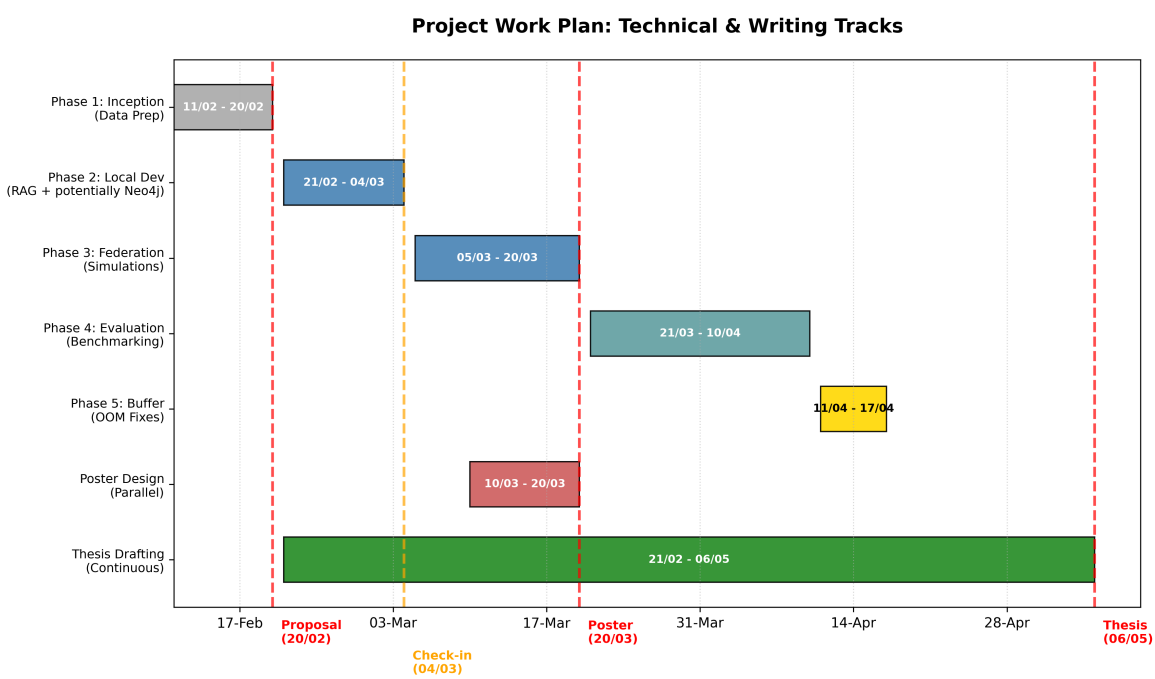

Figure 2: Project Gantt Chart showing parallel Technical and Writing tracks with buffer contingency.

3.2 Methodology

3.2.1 Simulation-First Architecture

The plan is structured around a "Simulation-First" methodology. Rather than attempting to deploy immediately to distributed hardware, Phase 2 focuses entirely on establishing a working **Local RAG** system. This isolates the complexity of "Reasoning" from the complexity of "Networking." Only once the model can successfully generate valid queries on a single node (Milestone M2) will the **Flower** federation layer be introduced in Phase 3. This dependency structure ensures that the Poster deadline (Mar 20) can be met with a working prototype, even if the federation experiments require further tuning during the evaluation phase.

3.2.2 Resource Plan

To ensure viability within the timeline, the project primarily relies on accessible, free-tier resources:

- **Compute:** Google Colab (Tesla T4 GPU, 16GB VRAM [11]) is sufficient for the target model (DeepSeek-R1-Distill-8B [6]) using 4-bit quantization (∼6GB VRAM load).
- **Software Stack:** The Flower framework's "Virtual Client Engine" allows for the serial simulation of multiple banks on a single GPU, removing the need for physical multi-node hardware [9].
- **Data Strategy:** The IBM AML Dataset is synthetic and open-source, eliminating ethical clearance delays associated with real banking data [8].

3.2.3 Contingency Planning

A dedicated Buffer Week (Apr 11 – 17) is allocated between the Evaluation and final Write-up phases. This contingency is specifically reserved for resolving "Out-Of-Memory" (OOM) errors that frequently occur when scaling from local execution to federated simulation. If Phase 3 runs over schedule, the scope of the Evaluation can be reduced by removing the "Centralized Control" comparison, ensuring the core objectives are still met by the deadline.

3.2.4 Risks and Mitigation

The following risks have been identified based on the constraints of the Google Colab environment and the experimental nature of "Reasoning" models.

- **Risk 1: Technical (Resource Limits) – Significance: High**
  The System RAM on Google Colab [12] is a hard bottleneck. Loading the full IBM transaction graph while simultaneously running the LLM may cause the session to crash (Out-Of-Memory).

  Mitigation: Utilise a low-resolution dataset. Use **4-bit Quantization (QLoRA)** to reduce the model's VRAM footprint to ∼6GB [13].

- **Risk 2: Methodological (Simulation Fidelity) – Significance: Medium**
  A single-machine simulation does not inherently capture the network latency, packet loss, or straggler behavior of a real-world distributed system.

  Mitigation: Introduce artificial time delays and Gaussian noise within the Flower aggregation loop. This mathematically models heterogeneous network conditions, ensuring the evaluation metrics reflect a realistic deployment scenario.

- **Risk 3: Model Reliability (AI Safety) – Significance: Medium**
  "Reasoning" models can hallucinate, producing invalid database interaction syntax (e.g., non-existent relationship types), which would crash the execution pipeline during automated testing.

  Mitigation: Evaluate the system against safety metrics. Implement a deterministic Syntax Validator (Linter) middleware. Invalid queries will be caught pre-execution and flagged as "Reasoning Failures" for the Utility metric, allowing the simulation to continue without crashing.

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